Chapter 9

BJT Configurations & Characteristics

9.1 Introduction

A transistor is a three-terminal device (Emitter, Base, Collector). However, any electronic circuit typically operates as a ”Two-Port Network” requiring four terminals: two for the Input port and two for the Output port.

To adapt the 3-terminal BJT into a 4-terminal network, one of the transistor’s terminals must be made **common** to both the input and output circuits (usually connected to ground). This gives rise to three fundamental circuit configurations:

1. **Common Base (CB):** Base is grounded.
2. **Common Emitter (CE):** Emitter is grounded.
3. **Common Collector (CC):** Collector is grounded.

Each configuration has unique characteristics regarding input impedance, output impedance, voltage gain, and current gain, making them suitable for specific applications.

Learning Objectives

By the end of this chapter, you should be able to:

- Sketch the circuit diagrams for CB, CE, and CC configurations.
- Interpret Input and Output characteristic curves for each configuration.
- Define Cutoff, Saturation, and Active regions on the CE output curve.
- Compare the Impedance (Z), Voltage Gain (A_v), and Current Gain (A_i) of the three modes.
- Explain the concept of Phase Shift in Common Emitter amplifiers.

9.2 Common Base (CB) Configuration

In this configuration, the **Base** is common to both input and output.

- **Input Signal:** Applied between Emitter and Base.
- **Output Signal:** Taken between Collector and Base.

9.2.1 Circuit Diagram

To operate in the active region:

- The Emitter-Base junction is **Forward Biased** (V_{EE}).
- The Collector-Base junction is **Reverse Biased** (V_{CC}).

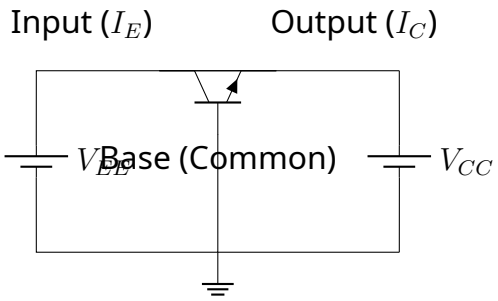


Figure 9.1: Common Base Configuration (NPN).

9.2.2 Characteristics

1. Input Characteristics (I_E vs V_{BE}) Since the input is a forward-biased diode (Base-Emitter), the graph resembles a standard diode curve.

- As V_{BE} exceeds 0.7V (Silicon), I_E increases rapidly.
- **Input Impedance (Z_{in}):** Very Low (typically $20\Omega - 50\Omega$).

2. Output Characteristics (I_C vs V_{CB})

- I_C is almost independent of V_{CB}. It depends entirely on I_E.
- The curves are flat, horizontal lines.
- **Output Impedance (Z_{out}):** Very High (typically $1M\Omega$).

3. Gains

- **Current Gain (α):** $I_C/I_E \approx 0.98$ (Less than 1).
- **Voltage Gain (A_v):** High.
- **Phase Shift:** 0° (Input and Output are in phase).

Applications: High-frequency applications (Radio Frequency amplifiers) because the low input capacitance allows it to work at high speeds.

9.3 Common Emitter (CE) Configuration

This is the most widely used configuration in electronics. The **Emitter** is common (usually grounded).

- **Input Signal:** Applied between Base and Emitter.
- **Output Signal:** Taken between Collector and Emitter.

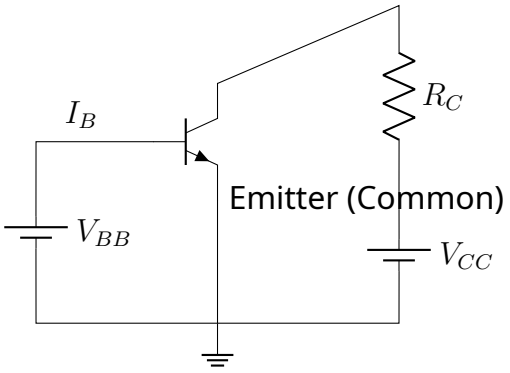


Figure 9.2: Common Emitter Configuration (NPN).

9.3.1 Circuit Diagram

9.3.2 Output Characteristics (I_C vs V_{CE})

This set of curves is the most important tool for transistor analysis. It plots the Collector Current (I_C) against the Collector-Emitter Voltage (V_{CE}) for various fixed values of Base Current (I_B).

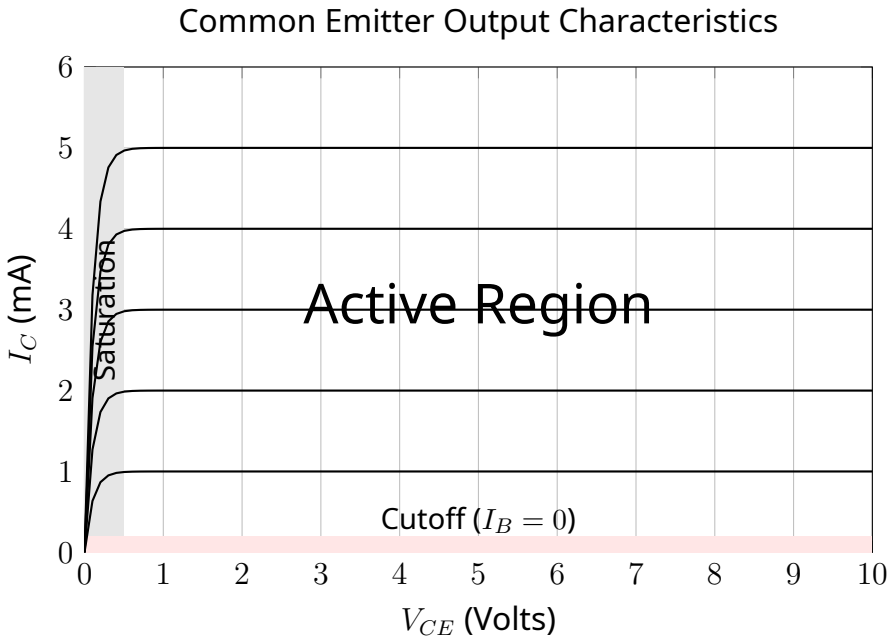


Figure 9.3: Output Characteristics of a CE Transistor.

The graph defines three distinct operating regions:

1. **Active Region:**

   - Located in the center. The B-E junction is forward biased, C-B is reverse biased.

- I_C is proportional to I_B ($I_C = \beta I_B$).
- Used for **Amplification**.

2. **Cutoff Region:**
   - Located at the bottom ($I_B = 0$). Both junctions are reverse biased.
   - $I_C \approx 0$ (Open Switch).
3. **Saturation Region:**
   - Located at the far left ($V_{CE} < 0.7$V). Both junctions are forward biased.
   - Current is maximum ($I_{C(sat)} \approx V_{CC}/R_C$).
   - Acts as a Closed Switch.

9.3.3 CE Parameters

- **Current Gain (β):** Large (typically 50–300).
- **Voltage Gain:** High.
- **Input Impedance:** Medium ($1k\Omega$).
- **Output Impedance:** Medium ($50k\Omega$).
- **Phase Shift:** 180°. The output signal is inverted relative to the input.

The Phase Inversion

In a CE amplifier, when the input voltage (V_{BE}) increases, I_B increases $\rightarrow I_C$ increases. The output voltage is taken across the transistor (V_{CE}).

$$V_{CE} = V_{CC} - I_C R_C$$

As I_C goes UP, the drop across R_C goes UP, so V_{CE} goes **DOWN**. Hence, an inverted output.

9.4 Common Collector (CC) Configuration

Also known as the **Emitter Follower**. The **Collector** is common (usually connected to V_{CC} which is an AC ground).

- **Input:** Base.
- **Output:** Emitter.

9.4.1 Circuit Diagram

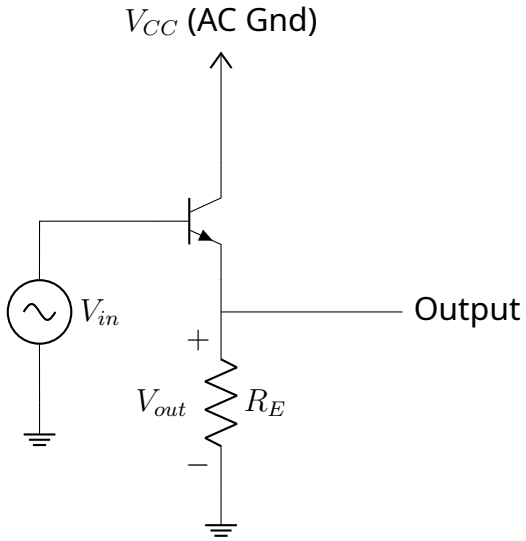


Figure 9.4: Common Collector (Emitter Follower) Circuit.

9.4.2 Characteristics

Why is it called an Emitter Follower? Because the output voltage at the Emitter "follows" the input voltage at the Base perfectly.

$$V_{out} \approx V_{in} - 0.7\text{V}$$

Key Parameters:

- **Voltage Gain:** Slightly less than 1 (Unity Gain).
- **Current Gain:** High ($\gamma = \beta + 1$).

- **Input Impedance:** Very High.
- **Output Impedance:** Very Low.
- **Phase Shift:** $0°$.

Application: Impedance Matching or **Buffer**. It connects a high-impedance source to a low-impedance load (like a speaker) without signal loss.

9.5 Comparison Summary

Table 9.1: Comparison of BJT Configurations

Parameter	Common Base	Common Emitter	Common Collector
Input/Output	E / C	B / C	B / E
Input Impedance	Low ($\approx 50\Omega$)	Medium ($\approx 1k\Omega$)	High ($\approx 300k\Omega$)
Output Impedance	High ($\approx 1M\Omega$)	Medium ($\approx 50k\Omega$)	Low ($\approx 50\Omega$)
Voltage Gain	High (≈ 150)	Medium (≈ 100)	Low (≈ 1)
Current Gain	Low (< 1)	High (β)	High ($\beta + 1$)
Power Gain	Medium	Highest	Medium
Phase Shift	$0°$	$180°$	$0°$
Typical Use	High Freq (RF)	Audio Amp	Impedance Matching

9.6 Chapter Summary

1. **Common Base:** Good for high frequency, voltage amplification, but no current gain.
2. **Common Emitter:** The ”workhorse” of electronics. Provides both voltage and current gain. Inverts the signal.
3. **Common Collector:** Acts as a buffer. High input Z, Low output Z. Output follows input.
4. **Output Characteristics:** Defined by Saturation (On), Cutoff (Off), and Active (Linear) regions.

9.7 Review Questions

Part A: Conceptual Questions

1. Explain the concept of ”Phase Reversal” in a Common Emitter amplifier. Why does it occur?
2. Why is the Common Collector configuration often referred to as an ”Emitter Follower”?
3. Which configuration would you select to connect a high-impedance guitar pickup to a low-impedance speaker? Explain your reasoning.
4. Define the ”Saturation Region” of a BJT. What are the bias conditions for both junctions in this state?
5. Why does the Common Base configuration have the lowest input impedance of all three types?
6. Sketch the output characteristics of a CE transistor. Label the three operating regions.
7. Which configuration provides the highest Power Gain? Why?
8. In the Saturation region, does the collector current depend on the base current? Explain.
9. What is the relationship between α and the current gain of a Common Base amplifier?
10. Why is the Common Emitter configuration the most commonly used for general-purpose amplification?

Part B: Advanced Calculations & Analysis

11. **Common Base Calculation:** A transistor in a Common Base configuration has an Emitter current $I_E = 4$ mA and an alpha $\alpha = 0.98$.
    (a) Calculate the Collector current I_C.

(b) Calculate the Base current I_B.

(c) Determine the value of β for this transistor.

12. **CE Analysis:** A Common Emitter amplifier has a base current $I_B = 25\mu$A. The transistor has $\beta = 120$.

(a) Calculate I_C.

(b) Calculate I_E.

(c) If the load resistor $R_C = 2k\Omega$ and $V_{CC} = 15$V, calculate the voltage drop across the load resistor and the value of V_{CE}.

13. **Power Gain Calculation:** A CE amplifier has an input resistance of $1k\Omega$ and an output load of $5k\Omega$. The current gain $\beta = 50$.

(a) Estimate the Voltage Gain ($A_v \approx \beta \frac{R_{out}}{R_{in}}$).

(b) Calculate the Power Gain ($A_p = A_v \times \beta$).

(c) Convert the Power Gain to decibels (dB).

14. **Common Collector (Buffer):** In a Common Collector circuit, the input base current changes by 10μA, causing the Emitter current to change by 1 mA.

(a) Calculate the current gain of this configuration.

(b) Determine the β value of the transistor.

(c) If the input voltage change was 0.5V, calculate the input impedance.

15. **Leakage Analysis:** A Germanium transistor in CE mode has $\beta = 50$ and $I_{CBO} = 2\mu$A.

(a) Calculate the collector current if $I_B = 0$ (Cutoff condition).

(b) Calculate the collector current if $I_B = 100\mu$A.

(c) What percentage of the total current in (b) is contributed by leakage?